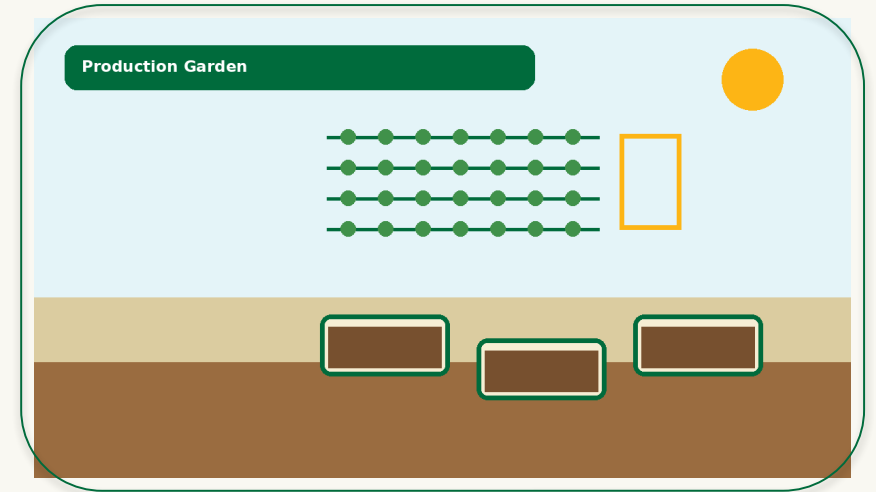


Production Garden

Applied Crop Production

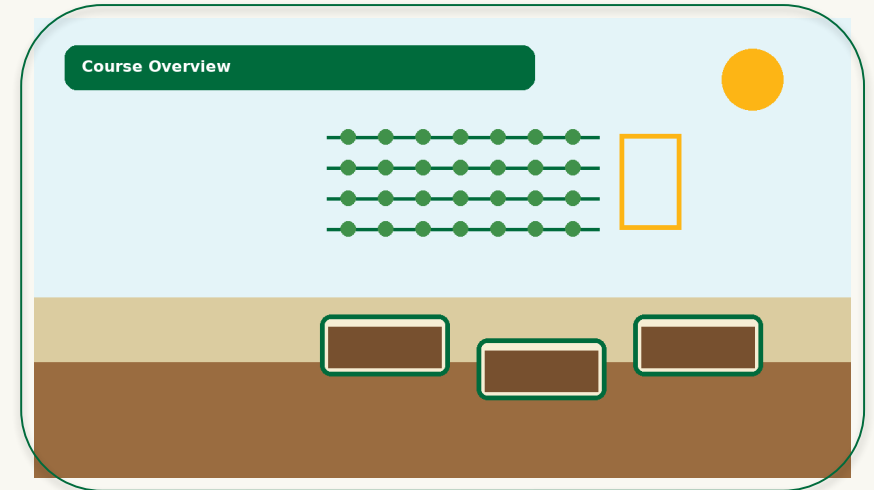
- LIFEWS AgriShine™ GrowMeal™ JSS 1-3 Teacher Training
- Audience: JSS 1-3 teachers, agricultural-science educators and garden facilitators
- Format: Self-paced online course + supervised practical/laboratory orientation
- Recommended duration: 3.5 hours online + 2 hours practical
- JSS learning progression: formulate -> test -> measure -> analyze -> manage



Course Overview

Learning outcomes, materials and teacher readiness

- Learning outcomes: explain the production or experimental system; teach all ten technical topics accurately; guide students in planning, measurement and data collection; connect agricultural science with mathematics and evidence; supervise field activities safely; maintain credible records; help students distinguish observation, analysis and conclusion.
- Core materials: teacher guide, plot/system map, crop plan, child-safe field tools, irrigation equipment, labels, clipboards, measuring tapes/rulers, scales and sensors where relevant, data sheets, clean harvest containers, PPE/handwashing supplies and first-aid access.
- Completion expectation: complete all modules, demonstrate one safe practical session, and submit a lesson, production plan or experimental protocol appropriate to the course.



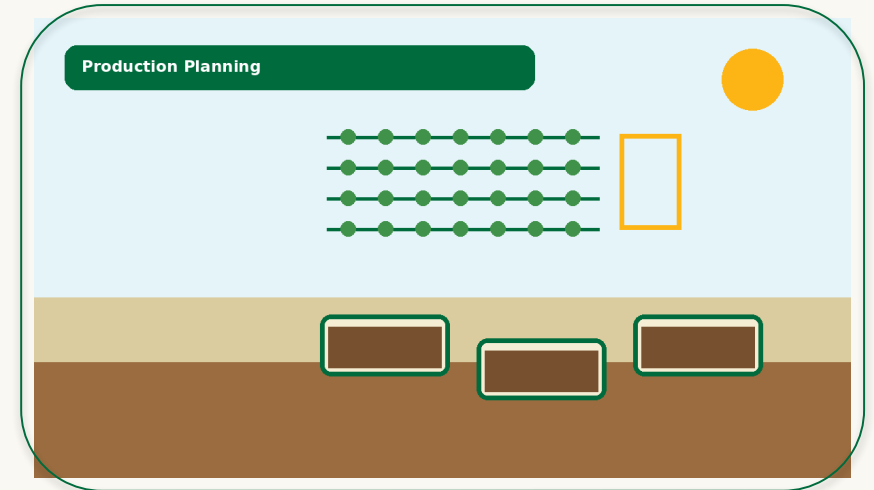
TEACHER PRACTICE

- Training sequence: QUESTION -> PLAN -> IMPLEMENT -> MEASURE -> ANALYZE -> DECIDE.

Production Planning

Teacher knowledge, technical practice and student supervision

- Define the production objective before planting: crop, intended use, available area, season and learning goals.
- Prepare a simple production calendar covering land preparation, planting, routine care, monitoring and expected harvest.
- Estimate required seed/seedlings, bed or plot area, water access and basic materials.
- Match the scale of production to the class's capacity to maintain it.
- Include contingency decisions for delayed planting, poor emergence or crop failure.



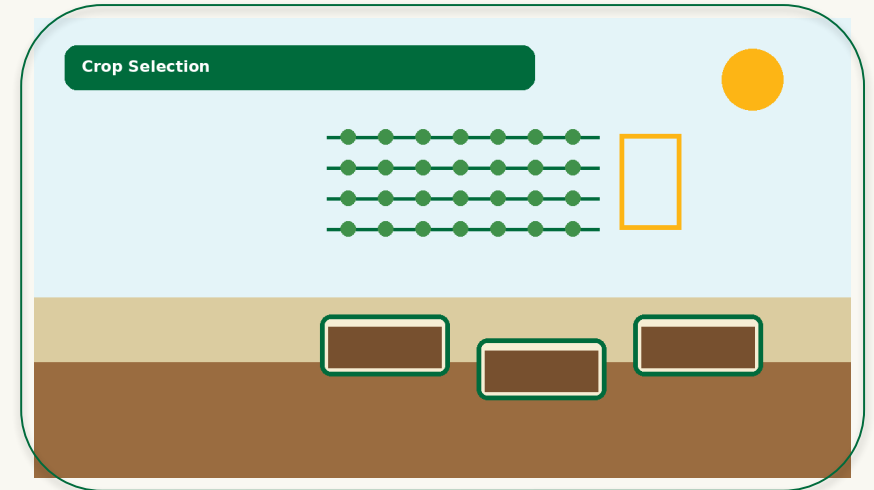
TEACHER PRACTICE

- Teacher task: prepare a one-term production plan for one crop.

Crop Selection

Teacher knowledge, technical practice and student supervision

- Select crops using season, site conditions, water availability, production duration, local relevance and school calendar.
- Compare varieties only when reliable information is available.
- Consider crop growth habit, pest pressure, harvest method and learning value.
- Avoid selecting crops solely because they are visually attractive.
- Document the reason for each crop choice.



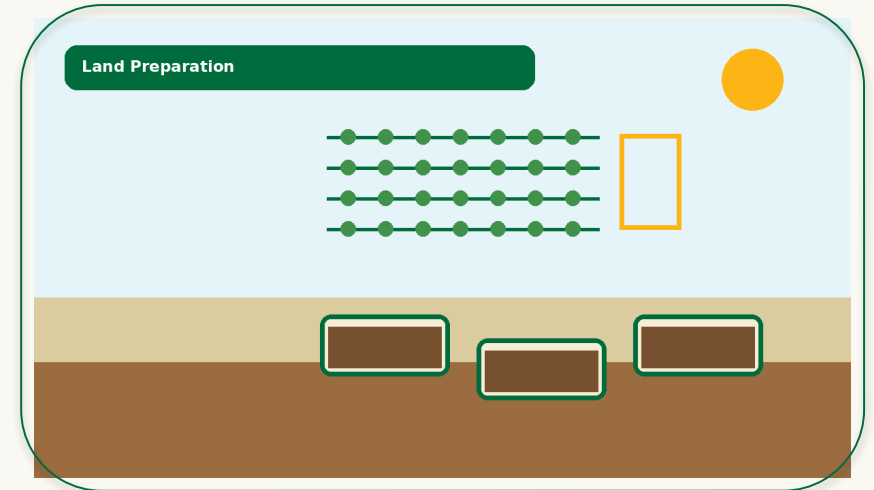
TEACHER PRACTICE

- Prepare a crop-selection matrix using at least four criteria.

Land Preparation

Teacher knowledge, technical practice and student supervision

- Assess the plot for drainage, previous use, weeds, compaction and access before cultivation.
- Define beds, rows and paths to reduce trampling and improve management.
- Use adult-led procedures for heavy digging and hazardous tools.
- Incorporate only mature, approved soil amendments.
- Record the preparation method so later performance can be interpreted.



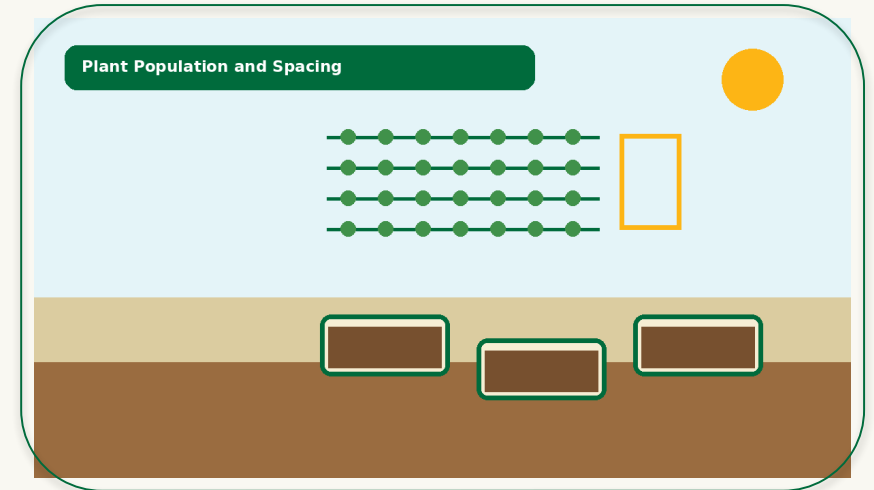
TEACHER PRACTICE

- Create a land-preparation checklist and plot sketch.

Plant Population and Spacing

Teacher knowledge, technical practice and student supervision

- Define plant population as the number of established plants in a known area.
- Use recommended within-row and between-row spacing for the selected crop.
- Calculate expected population from plot dimensions and spacing, then compare with actual establishment.
- Explain how excessive or inadequate density can influence competition, access and yield.
- Record replanting or thinning that changes population.



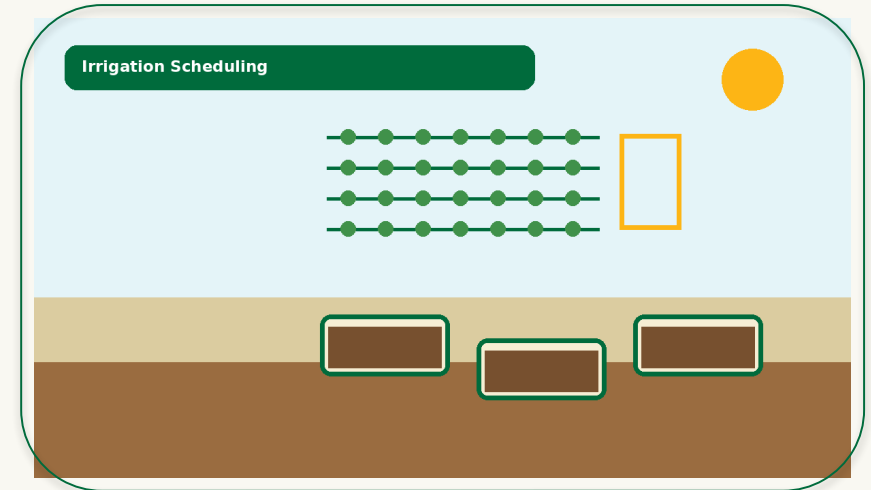
TEACHER PRACTICE

- Calculate target and actual plant population for one plot.

Irrigation Scheduling

Teacher knowledge, technical practice and student supervision

- Base irrigation timing on crop stage, soil moisture, weather conditions and system capacity rather than a fixed clock alone.
- Distinguish irrigation frequency from irrigation amount.
- Use soil-moisture observations or sensors where available.
- Record irrigation dates and estimated/measured water applied.
- Avoid deliberate severe water stress in school food-production plots.



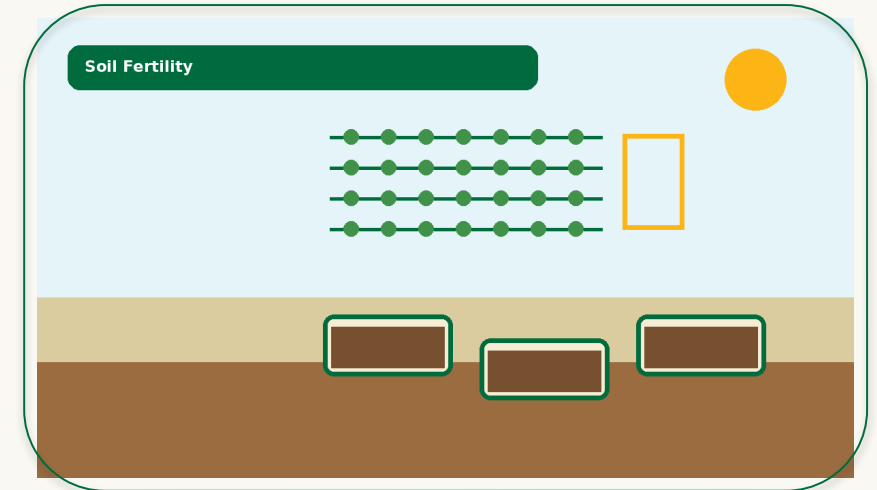
TEACHER PRACTICE

- Prepare a one-week irrigation decision sheet.

Soil Fertility

Teacher knowledge, technical practice and student supervision

- Review the crop's nutrient requirements and the garden's approved fertility plan.
- Use mature compost and approved fertilizers at appropriate rates under adult supervision.
- Explain macro-nutrients conceptually without diagnosing deficiency from appearance alone.
- Record amendment type, amount, date and plot.
- Use soil testing where available to improve decisions.



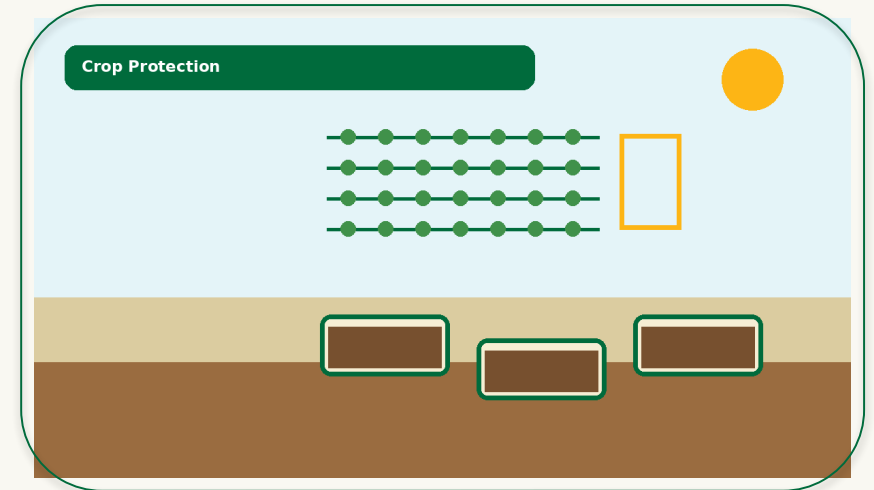
TEACHER PRACTICE

- Prepare a fertility input record for one production plot.

Crop Protection

Teacher knowledge, technical practice and student supervision

- Use regular scouting to identify weeds, pest damage, disease symptoms and physical injury.
- Apply integrated pest-management logic: prevention, observation, correct identification and proportionate response.
- Do not allow pupils to handle pesticides or hazardous control products.
- Distinguish beneficial organisms from crop pests.
- Record problem location, severity and action taken.



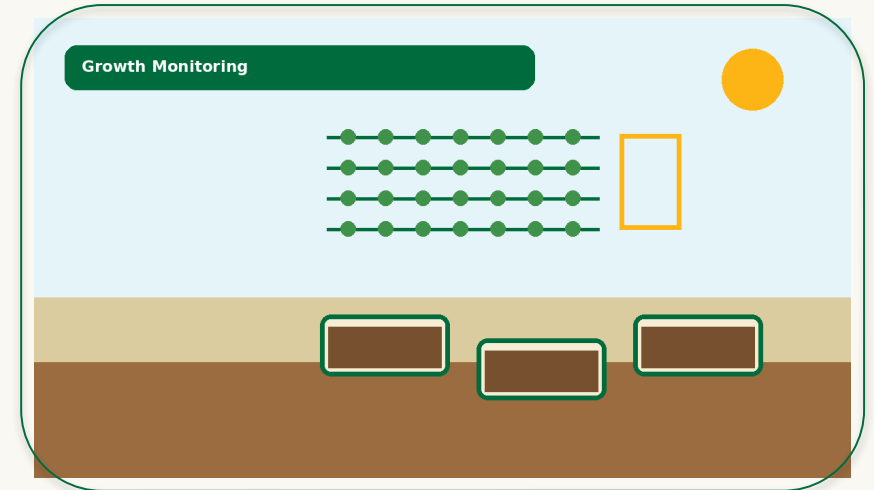
TEACHER PRACTICE

- Develop a weekly crop-scouting routine.

Growth Monitoring

Teacher knowledge, technical practice and student supervision

- Select indicators such as height, leaf number, canopy width, flowering or survival rate.
- Use marked sample plants and a consistent measurement protocol.
- Collect measurements at defined intervals.
- Record environmental or management events that may affect interpretation.
- Use graphs to visualize change over time.



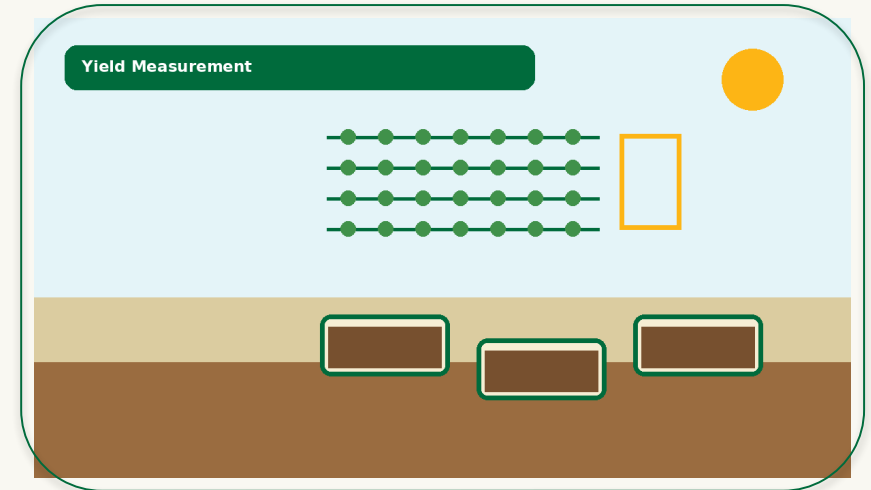
TEACHER PRACTICE

- Prepare a monitoring protocol for ten sample plants.

Yield Measurement

Teacher knowledge, technical practice and student supervision

- Define harvestable yield using a known plot area and consistent harvest criteria.
- Measure count and/or mass using calibrated or checked equipment.
- Record marketable and damaged produce separately when relevant to the learning goal.
- Calculate simple yield per plot or per unit area.
- Avoid attributing yield differences to one factor without supporting evidence.



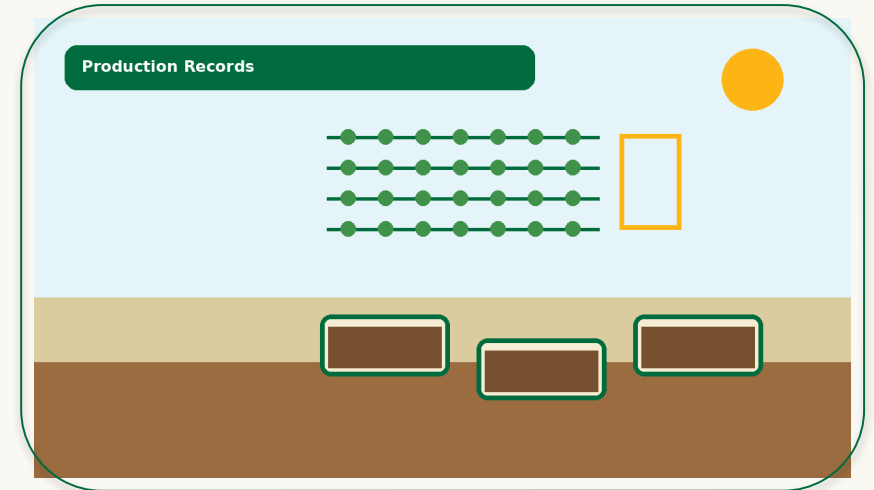
TEACHER PRACTICE

- Prepare a yield-measurement and calculation exercise.

Production Records

Teacher knowledge, technical practice and student supervision

- Maintain plot identity, crop/variety, dates, inputs, irrigation, crop-protection observations, growth data and harvest records.
- Use consistent units and standardized forms.
- Correct errors transparently rather than deleting inconvenient data.
- Use records to evaluate management decisions after harvest.
- Archive records for comparison across terms.



TEACHER PRACTICE

- Completion task: build a complete production record and 45-minute teaching activity.